

MICROMOLD · REQUIREMENTS

MicroMold requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Status from MMD-CAL-001 v0.4 for the constructable design (MMD-DDR-003); R14 measured against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from MMD-CAL-001; R14 redefined to the press excluding molds and R13 to name the hood (MMD-DDR-001 D8, D9)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against first-order estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

MicroMold requirements

These are the requirements for the concept, checked by calculation at TRL 3 in MMD-CAL-001. Targets are proposals for review, not yet validated with users, and will be revised after co-design sessions (see MMD-PRB-001). "Met on paper" means met by calculation, and "met by design" means met by a stated feature; neither is demonstrated on hardware. Two targets changed in v0.3 under MMD-DDR-001: R14 covers the press without molds (D8), and R13 names the hood (D9). Three more changed in v0.4 under MMD-DDR-002, decided by Amish on 2026-09-25: R1 requires injection-grade flake, R9 allows 40 kg (was 35 kg) and R14 allows \$500 (was \$400). In v0.5 R14 allows \$520, the budget Amish approved on 2026-09-26 to cover the priced BOM (MMD-DDR-002). In v0.6 that figure is read as a hypothetical value-engineering target, not a limit (Amish, 2026-10-01), and status is from MMD-CAL-001 v0.4, for the constructable design of MMD-DDR-003.

The **reference part** used throughout is a 30 g HDPE test plaque, 64 x 50 x 6 mm, molded from washed, dried flake of 3 to 8 mm.

Table 1. Requirements.

ID	Requirement	Target	Status (MMD-CAL-001)	Verification (TRL 3 or later)
R1	Process the safe recycled thermoplastics	HDPE, PP, LDPE and PS flake of 3 to 8 mm from injection-molded items (caps, crates, buckets); bottle-grade HDPE and PVC excluded by procedure and labeling	Met by design (bottle grade would need about 24 MPa, CAL E6)	Design review; material trials later
R2	Shot size	30 g or more of HDPE per injection	At risk: 34.2 g ideal, about 25 g if the fresh charge is not tamped	Swept-volume calculation; later weighed shots
R3	Injection pressure	8 MPa (80 bar) or more at the melt with 250 N or less on the handle	Met on paper: 8.9 MPa at 0.60 drive efficiency (12.4 MPa at 0.84); fill needs about 4.8 MPa	Force and friction calculation; later load cell reading
R4	Temperature control	Barrel and nozzle zones each settable from 150 to 260 °C, held within ± 5 °C at the thermocouple; independent cut-out above 280 °C	Met by design, unverified	Controller and heater datasheets; later logging
R5	Warm-up	Cold start to 220 °C in 15 min or less	Met on paper: 12.3 min with two 300 W bands	Heat capacity calculation; later timed start
R6	Throughput	8 or more reference parts per hour with one mold	Met on paper: 11.6 per hour with the mold cooling fan, mold about 47 °C (soak limited)	Cycle-time estimate; later timed trials
R7	Mold envelope	Molds up to 150 x 120 mm footprint and 40 to 120 mm stack height; 40 cm ² or more projected area at full pressure	Met on paper: 45 cm ² ; stacks of 30 to 120 mm fit	Clamp force calculation; model
R8	Low-cost tooling	A two-plate aluminum mold for the reference part machinable on a manual mill or small CNC for \$100 or less	Met on paper: \$94 (indicative)	Quotation from a local shop
R9	Bench size and mass	Press footprint 350 x 300 mm or less, top of handle 1.1 m or less above the bench, mass 40 kg or less without the control box	At risk on mass: 39.0 kg with an assumed 8 kg press head; 320 x 260 mm and 1,094 mm met	Model and mass estimate
R10	Power supply	Single-phase 230 V or 120 V, 1 kW or less, 10 A or less at 120 V	Met on paper: 753 W, 6.3 A at 120 V	Heater and fan ratings
R11	Touch-safe outer surfaces	Guard and jacket 60 °C or less at 220 °C set point and 25 °C ambient; nozzle and mold zone guarded	Met on paper: skin 48 °C; nozzle zone shield added	Thermal calculation
R12	Electrical safety	Earthed frame, fused inlet, double-pole switch, RCD or GFCI supply, heaters and wiring rated for 250 °C at the barrel	Met by design, unverified	Design review against IEC 60204-1 principles
R13	Fume control	A side hood with a duct fan at the funnel, and operation only under it or outdoors;	Met on paper: about 124 m ³ /h	Hood calculation; design review

ID	Requirement	Target	Status (MMD-CAL-001)	Verification (TRL 3 or later)
		set points above 260 °C blocked in the controller		
R14	Affordable	Parts for the press, excluding molds (tooling, see R8), at or under the \$520 value-engineering target (a hypothetical control target)	Over the value-engineering target: \$544, USD 24 over (\$638 with one mold)	Priced BOM (bom/ bom.csv)
R15	Garage-buildable	Everything except the barrel and the mold built with hand tools, a drill press and optional welding	Met by design: the barrel with its nozzle and the molds come from a local machine shop; the plunger is only cross-drilled	Build sequence review
R16	Repeatable parts	Part mass within ± 3 % over 10 consecutive shots	Not verifiable at TRL 3	Later bench trials (TRL 4, not in the current phase)

■ Assumptions

- HDPE melt density about 0.75 g/cm³ at 200 to 220 °C; solid density about 0.95 g/cm³ (typical handbook values, estimates). The other property values are in MMD-CAL-001, Table 1.
- Arbor press drive: pinion pitch radius about 20 mm, 450 mm ratchet handle, design efficiency 0.60 (0.84 calculated), to be checked on the chosen press.
- Processing temperatures follow published melt ranges: HDPE 190 to 230 °C, PP 200 to 240 °C, LDPE 220 to 245 °C, PS 180 to 260 °C ([RJC Mold](#)).
- Four M10 class 8.8 cap screws in thread inserts, preloaded to about 20 kN each, clamp the mold plates, with a factor of 2 against melt pressure opening the parting plane.

■ Requirements not met or at risk

- **R14 cost:** value-engineering target: USD 520. Estimated cost of the constructable design: USD 544 for the press (USD 24 over the target), with molds counted as tooling; USD 638 with the first mold. The parts added to make the design buildable account for USD 37 (MMD-DDR-003); cost drivers and savings are in MMD-DEC-001.
- **R9 mass:** 39.0 kg against 40 kg, with an assumed 8 kg arbor press head; the margin depends on the real press.
- **R2 shot size:** met only if each top-up of flake is tamped with the press while loading.
- **R16** cannot be assessed until hardware exists (TRL 4, on hold).